1. Project Planning & Management

Project: NutriVision — AI-Driven Dietary Health Assistant
Program: DEPI, Round 5 — Cross-Platform / Flutter Track
Document status: Initial draft for team and instructor review
Instructor: Rana Mohy

1.1 Project Proposal

Background

Many users want to eat more healthily but find it difficult to understand the nutritional value of everyday meals, maintain a consistent meal record, and find practical alternatives that match their preferences or dietary restrictions.

Proposed solution

NutriVision is a cross-platform mobile application that analyzes meal descriptions and provides AI-assisted nutritional guidance, healthier alternatives, and personalized recipe suggestions. It will also help users keep a meal history and view nutritional insights in a simple, responsive interface.

NutriVision provides guidance and educational information. It is not a replacement for a qualified doctor or registered dietitian, especially for medical conditions, allergies, pregnancy, or clinical meal plans.

Vision

To make everyday dietary decisions clearer, more personalized, and easier to maintain.

Objectives

1. Help users understand the nutritional characteristics of their meals.
2. Provide practical alternatives for less healthy meals and snacks.
3. Generate recipe ideas based on dietary restrictions and user preferences.
4. Give users a consistent way to record and review their meals.
5. Demonstrate a secure, testable, and maintainable Flutter application.

Scope

In scope for the initial release

- User profile and dietary preferences.
- Meal description and meal-log entry.
- Nutritional analysis request and result display.
- AI-assisted recipe and healthy-alternative suggestions.
- Meal history and basic nutritional charts.
- Real-time synchronization of meal logs with Firebase Cloud Firestore.
- Dio-based networking with interceptors for request/error handling.
- Runtime permission handling only where a confirmed feature requires it.
- Unit tests for calculation/business logic and widget tests for key screens.

Out of scope unless approved later

- Medical diagnosis or treatment recommendations.
- Automatic medical prescriptions or personalized clinical diets.
- Direct ordering or delivery of food.

- Full barcode-scanning or image-recognition features.
- Wearable-device integrations.
- A nutritionist consultation marketplace.

Target users

- People who want simple guidance for everyday meals.
- Users following dietary preferences or restrictions.
- Students and busy users who need quick meal ideas.
- Users who want to review meal history and improve consistency.

1.2 Project Plan

The schedule below is a relative plan. Exact dates will be added after the DEPI submission and presentation dates are confirmed.

Project timeline

Phase	Planned period	Main output
Planning	12–25 Oct 2026	Proposal, scope, and requirements
Design	26 Oct–08 Nov 2026	User flows, wireframes, and UI system
Development	09–29 Nov 2026	Flutter foundation, core flows, API, and Firestore
Quality	30 Nov–20 Dec 2026	Tests, bug fixing, documentation, and presentation

Milestones and deliverables

Milestone	Deliverable	Acceptance indicator
M1 — Scope approved	Project proposal and scope	Team and instructor feedback recorded
M2 — Requirements approved	Requirements document and use cases	Functional and non-functional requirements reviewed
M3 — Design approved	User flows, personas, and wireframes	Main user journeys are clear before implementation
M4 — Foundation ready	Flutter structure and navigation	App runs on the agreed target platform(s)
M5 — Core flow ready	Meal entry, analysis result, and history	Main MVP journey works with loading and error states
M6 — Integration ready	API and Firestore integration	Data handling and security rules are tested
M7 — Quality ready	Tests and bug-fix report	Critical tests pass and known issues are documented
M8 — Final delivery	Documentation and presentation package	Demo, README, report, and presentation are ready

1.3 Task Assignment and Roles

The following is a proposed allocation for discussion. It is not final until the team confirms it.

Team member	Proposed responsibility	Main outputs
Mohamed Samy	Team coordination, architecture, Git workflow, and integration	Task tracking, branch coordination, architecture decisions, and releases
Hager Awadeen	Frontend implementation	Screens, reusable widgets, responsive layouts, and user interactions
Menna Khalil	UX/UI design	User flows, wireframes, design system, and high-fidelity UI designs
Hoda Mohamed	Application logic	Business rules, validation, state management, and nutritional calculations
Nancy Ahmed	API, Firebase, and database integration	Dio services, interceptors, Firebase/Firestore setup, and data synchronization

Collaboration rules

- Work on a focused branch; do not push feature work directly to main.
- Use short, clear commit messages that describe the change.
- Open a pull request for review before merging into main.
- Run formatting, analysis, and tests before requesting review.
- Record decisions that affect the architecture, scope, or user experience.

1.4 Risk Assessment and Mitigation Plan

Risk	Probability	Impact	Mitigation
AI output is inaccurate or unsuitable	Medium	High	Use clear prompts, validate response structure, show a health disclaimer, and avoid diagnosis/treatment claims
External API is unavailable or slow	Medium	High	Add timeouts, loading/error states, safe fallbacks, and mock data for development
Firestore permissions expose user data	Low/Medium	High	Use Firebase Authentication, least-privilege Security Rules, and test unauthorized access
Scope grows beyond the delivery window	High	High	Freeze an MVP scope, prioritize must-have requirements, and track change requests
Merge conflicts or inconsistent code	Medium	Medium	Use focused branches, frequent pulls, formatting, and pull-request review
Team member availability changes	Medium	Medium	Document ownership, share knowledge, and keep tasks small enough to transfer
Sensitive keys are committed accidentally	Low	High	Use environment/local configuration, .gitignore, secret rotation, and repository checks
Nutrition calculations are incorrect	Medium	High	Define formulas, unit-test edge cases, and compare results against trusted references
UI is difficult to use on different screens	Medium	Medium	Use responsive layouts, test representative sizes, and perform a usability review

1.5 Key Performance Indicators (KPIs)

These targets are initial proposals and should be adjusted after the MVP is tested.

KPI	Initial target	Measurement method
Core-flow completion	At least 90% of test users complete meal entry and result viewing	Usability test checklist
Response experience	Initial feedback appears within 3 seconds where the service responds within that time; longer requests show progress	App logs and manual test runs
Error handling	All tested network and empty-response cases show a clear recoverable state	Test-case results
Reliability	No unresolved blocker or critical bug at final delivery	Issue tracker and release checklist
Test quality	All critical business rules have passing unit tests; key screens have widget tests	Test report
Data protection	No secrets in Git and unauthorized Firestore access is rejected	Repository audit and security-rule tests
Usability	At least 4 out of 5 in the team usability review	Structured review form

1.6 Open Decisions

- Final submission date and presentation date: TBD.
- Final AI provider and nutrition-data source: TBD.
- Authentication method: TBD.
- Final MVP screen list: TBD after wireframes.
- Exact team responsibilities: confirm with all members.